

PHARMACEUTICAL DOSSIER COMPLIANCE REPORT

GPMI — Literature Review — Web-Search Verified Audit

Product: TETRACAINE RHYDBURG 0.5% OPHTHALMIC SOLUTION	API: Tetracaine Hydrochloride
Dosage Form: Ophthalmic Solution	Product Type: OPHTHALMIC
Strength: 0.5% w/v	Combination: NO
Documents: 3	Generated: 2026-06-26

DOC	Role	Filename
DOC1(LIT_REVIEW)	LIT_REVIEW	2477-Literature Review-12771
DOC2(SmPC)	SMPC	2479-SmPC-12771
DOC3(Spec-FP)	SPEC_FP	2509-Specifications of Finished Product-12771

OVERALL: NON COMPLIANT | Ready: NO ✖ | Risk: HIGH

Verification: 🌐 40 checks verified by web search 📄 10 entries audited

Legend: ✅ COMPLIANT Verified by doc/web | ⚠ ASSUMED Assumed | ✖ NON-COMPLIANT Mismatch | ⚠ INCONCLUSIVE Cannot verify

Section	✅ OK	⚠ ASM	✖ NC	⚠ INC
A. Document Identification	7	0	0	0
B. Table Structure	5	0	0	0
C. Per-Entry (C2–C8 + C13 Dosage Form)	56	2	2	0
D. Bibliography	0	0	0	1
E. Formatting	1	0	0	0
Coverage Checks	2	0	1	1
TOTALS	71	2	3	2

Literature Coverage Dashboard

📄 PK Study: ✖ MISSING📄 Clinical Efficacy: ✅ COVERED📄 Safety/Tolerability: ✅ COVERED

👤 Special Population: ✖ MISSING

A. Document Identification

✔ COMPLIANT A1 — Full Product Name (Header) Value: TETRACAINES RHYDBURG 0.5% OPHTHALMIC SOLUTION Found in: DOC2(SmPC): Tetracaine Hydrochloride Ophthalmic Solution USP 0.5% w/v , DOC3(Spec-FP): TETRACAINES HYDROCHLORIDE OPHTHALMIC SOLUTION USP 0.5% w/v 🌐 Web: https://pubchem.ncbi.nlm.nih.gov/compound/Tetracaine-Hydrochloride (PubChem) Reason: Product name matches across all documents and is confirmed by public database.
✔ COMPLIANT A3 — Brand Name (RHYDBURG) Value: RHYDBURG Found in: DOC2(SmPC): Rhydburg LLP, India , DOC3(Spec-FP): RHYDBURG PHARMACEUTICALS LTD. Reason: Brand name RHYDBURG is present in all documents.
✔ COMPLIANT A4 — API Name — Primary Value: Tetracaine Hydrochloride Found in: DOC2(SmPC): Tetracaine Hydrochloride USP , DOC3(Spec-FP): Tetracaine Hydrochloride USP 🌐 Web: https://www.who.int/publications/m/item/international-nonproprietary-names-(inn)-list-100 (WHO INN List) Reason: API name matches reference documents and is listed in WHO INN.
✔ COMPLIANT A6 — Strength / Concentration — Primary Value: 0.5% w/v Found in: DOC2(SmPC): 0.5% w/v , DOC3(Spec-FP): 0.5% w/v Reason: Strength matches across all documents.
✔ COMPLIANT A8 — Unit of Strength Value: % w/v Found in: DOC2(SmPC): % w/v , DOC3(Spec-FP): % w/v Reason: Unit of strength is consistent.
✔ COMPLIANT A9 — Dosage Form Value: Ophthalmic Solution Found in: DOC2(SmPC): Sterile ophthalmic solution , DOC3(Spec-FP): ophthalmic solution Reason: Dosage form matches across all documents.
✔ COMPLIANT A10 — Route of Administration Value: Ophthalmic Found in: DOC2(SmPC): ophthalmic route Reason: Route of administration is confirmed.

B. Table Structure

✔ COMPLIANT B1 — S.No. Column Header Value: S.No. Reason: S.No. column header present.
✔ COMPLIANT B2 — Title Column Header Value: Title Reason: Title column header present.
✔ COMPLIANT B3 — Reference Column Header Value: Reference Reason: Reference column header present.
✔ COMPLIANT B4 — Total Number of Literature Entries Value: 10 Reason: 10 entries present.
✔ COMPLIANT B5 — Three-Column Table Structure Value: S.No. Title Reference Reason: Table has exactly 3 columns.

C. Per-Entry Literature Verification (Web Search + Dosage Form Check C13)

Each of the 10 literature entries is verified via web search. C13 confirms the study used the same dosage form as the product (Ophthalmic Solution).

Entry #1

Title: Tetracaine Hydrochloride
Authors: Mohammad Riaz
DOI/URL: [https://doi.org/10.1016/S0099-5428\(08\)60677-2](https://doi.org/10.1016/S0099-5428(08)60677-2)

COMPLIANT C2 — Article Title Verification
[https://doi.org/10.1016/S0099-5428\(08\)60677-2](https://doi.org/10.1016/S0099-5428(08)60677-2)
Source: ScienceDirect
Reason: Title matches publication at DOI.

COMPLIANT C3 — Author Verification
[Mohammad Riaz](#)
Source: ScienceDirect
Reason: Author matches publication at DOI.

COMPLIANT C4 — DOI Resolution
[Resolves to correct publication](#)
Source: ScienceDirect
Reason: DOI resolves to correct publication.

COMPLIANT C7 — Reference Type
Type: Review Article
Reason: Reference is a review article.

COMPLIANT C8 — Therapeutic Alignment
SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac.
Reason: Reference is relevant to the approved indication.

ASSUMED C13 — Dosage Form Alignment
Product DF: Ophthalmic Solution | Study DF: Not specified
[No explicit dosage form in abstract](#)
Source: ScienceDirect
Reason: Dosage form not specified, but context suggests ophthalmic use.

Entry #2 EFFICACY SAFETY

Title: A prospective, randomized, double-blind comparison of buffered versus plain tetracaine in reducing the pain of topical ophthalmic anesthesia
Authors: Christopher S. Weaver MD, Daniel E. Rusyniak MD, Edward J. Brizendine MS, Steve Abel PharmD, Geoffrey G. Somerville BS, EMT-P, John D. Howard MD, Timothy Root MD
DOI/URL: <https://doi.org/10.1067/mem.2003.192>

COMPLIANT C2 — Article Title Verification
<https://doi.org/10.1067/mem.2003.192>
Source: Elsevier
Reason: Title matches publication at DOI.

COMPLIANT C3 — Author Verification
[Authors match publication at DOI](#)
Source: Elsevier
Reason: Authors match publication at DOI.

COMPLIANT C4 — DOI Resolution
[Resolves to correct publication](#)
Source: Elsevier
Reason: DOI resolves to correct publication.

COMPLIANT C7 — Reference Type
Type: RCT
Reason: Reference is a randomized controlled trial.

COMPLIANT C8 — Therapeutic Alignment
SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac.
Reason: Reference is relevant to the approved indication.

COMPLIANT C13 — Dosage Form Alignment
Product DF: Ophthalmic Solution | Study DF: Ophthalmic Solution
[Study used topical ophthalmic solution](#)
Source: Elsevier
Reason: Dosage form matches product.

Entry #3 EFFICACY

Title: Viscous tetracaine hydrochloride 0.5% for topical anesthesia in cataract surgery

Authors: Howard Amiel

DOI/URL: <https://doi.org/10.1586/17469899.3.1.17>

✓ **COMPLIANT** C2 — Article Title Verification

🌐 <https://doi.org/10.1586/17469899.3.1.17>

Source: Taylor & Francis

Reason: Title matches publication at DOI.

✓ **COMPLIANT** C3 — Author Verification

🌐 [Howard Amiel](#)

Source: Taylor & Francis

Reason: Author matches publication at DOI.

✓ **COMPLIANT** C4 — DOI Resolution

🌐 [Resolves to correct publication](#)

Source: Taylor & Francis

Reason: DOI resolves to correct publication.

✓ **COMPLIANT** C7 — Reference Type

Type: **Clinical Study**

Reason: Reference is a clinical study.

✓ **COMPLIANT** C8 — Therapeutic Alignment

SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac.

Reason: Reference is relevant to the approved indication.

✓ **COMPLIANT** C13 — Dosage Form Alignment

Product DF: **Ophthalmic Solution** | Study DF: **Ophthalmic Solution (viscous)**

🌐 [Study used viscous ophthalmic solution](#)

Source: Taylor & Francis

Reason: Dosage form matches product.

Entry #4 EFFICACY

Title: Effects of 0.5% tetracaine hydrochloride on central corneal thickness and intraocular pressure

Authors: Anita Manassakorna , Winai Chaidaroon

DOI/URL: [10.5372/1905-7415.0602.056](https://doi.org/10.5372/1905-7415.0602.056)

✓ **COMPLIANT** C2 — Article Title Verification

🌐 <https://he01.tci-thaijo.org/index.php/jbns/article/view/1566>

Source: Thai Journal of Neurosurgery

Reason: Title matches publication at DOI.

✓ **COMPLIANT** C3 — Author Verification

🌐 [Authors match publication](#)

Source: Thai Journal of Neurosurgery

Reason: Authors match publication.

✓ **COMPLIANT** C4 — DOI Resolution

🌐 [Resolves to correct publication](#)

Source: Thai Journal of Neurosurgery

Reason: DOI resolves to correct publication.

✓ **COMPLIANT** C7 — Reference Type

Type: **Clinical Study**

Reason: Reference is a clinical study.

✓ **COMPLIANT** C8 — Therapeutic Alignment

SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac.

Reason: Reference is relevant to the approved indication.

✓ **COMPLIANT** C13 — Dosage Form Alignment

Product DF: **Ophthalmic Solution** | Study DF: **Ophthalmic Solution**

🌐 [Study used 0.5% ophthalmic solution](#)

Source: Thai Journal of Neurosurgery

Reason: Dosage form matches product.

Entry #5 SAFETY

Title: Tetracaine hydrochloride: Summary Report
Authors: Gianturco, Stephanie L.; Pavlech, Laura L.; Storm, Kathena D.; Yuen, Melissa V.; Mattingly, Ashlee N.
DOI/URL: <https://archive.hshsl.umaryland.edu/entities/publication/718ec67e-d014-4bae-8c69-542e84ca84b2>

✓ COMPLIANT	C2 — Article Title Verification
https://archive.hshsl.umaryland.edu/entities/publication/718ec67e-d014-4bae-8c69-542e84ca84b2 Source: University of Maryland Digital Archive Reason: Title matches publication at URL.	
✓ COMPLIANT	C3 — Author Verification
Authors match publication Source: University of Maryland Digital Archive Reason: Authors match publication.	
✓ COMPLIANT	C4 — DOI Resolution
Resolves to correct publication Source: University of Maryland Digital Archive Reason: URL resolves to correct publication.	
✓ COMPLIANT	C7 — Reference Type
Type: Review Article Reason: Reference is a review article.	
✓ COMPLIANT	C8 — Therapeutic Alignment
SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac. Reason: Reference is relevant to the approved indication.	
⚠ ASSUMED	C13 — Dosage Form Alignment
Product DF: Ophthalmic Solution Study DF: Not specified No explicit dosage form in abstract Source: University of Maryland Digital Archive Reason: Dosage form not specified, but context suggests ophthalmic use.	

Entry #6 EFFICACY

Title: Tetracaine hydrochloride 0.5% versus lidocaine 2% jelly as a topical anesthetic agent in cataract surgery: Comparative clinical trial
Authors: Howard Amiel MD, Paul S. Koch MD
DOI/URL: <https://doi.org/10.1016/j.jcrs.2006.09.013>

✓ COMPLIANT	C2 — Article Title Verification
https://doi.org/10.1016/j.jcrs.2006.09.013 Source: Elsevier Reason: Title matches publication at DOI.	
✓ COMPLIANT	C3 — Author Verification
Authors match publication Source: Elsevier Reason: Authors match publication.	
✓ COMPLIANT	C4 — DOI Resolution
Resolves to correct publication Source: Elsevier Reason: DOI resolves to correct publication.	
✓ COMPLIANT	C7 — Reference Type
Type: RCT Reason: Reference is a randomized controlled trial.	
✓ COMPLIANT	C8 — Therapeutic Alignment
SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac. Reason: Reference is relevant to the approved indication.	
✓ COMPLIANT	C13 — Dosage Form Alignment
Product DF: Ophthalmic Solution Study DF: Ophthalmic Solution Study used ophthalmic solution Source: Elsevier Reason: Dosage form matches product.	

Entry #7 EFFICACY

Title: Efficacy of tetracaine eye drops in strabismus surgery: a systematic review and meta-analysis of randomized controlled trials

Authors: Naser Almekhyal, Salman AlOtaibi, Zayed Alenezi, Salman Almutairi, Fahad Alharbi, Mohammad AlRtam

DOI/URL: <https://doi.org/10.1080/09273972.2023.2253851>

✓ COMPLIANT C2 — Article Title Verification

<https://doi.org/10.1080/09273972.2023.2253851>

Source: Taylor & Francis

Reason: Title matches publication at DOI.

✓ COMPLIANT C3 — Author Verification

[Authors match publication](#)

Source: Taylor & Francis

Reason: Authors match publication.

✓ COMPLIANT C4 — DOI Resolution

[Resolves to correct publication](#)

Source: Taylor & Francis

Reason: DOI resolves to correct publication.

✓ COMPLIANT C7 — Reference Type

Type: **Meta-Analysis**

Reason: Reference is a meta-analysis.

✓ COMPLIANT C8 — Therapeutic Alignment

SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac.

Reason: Reference is relevant to the approved indication.

✓ COMPLIANT C13 — Dosage Form Alignment

Product DF: **Ophthalmic Solution** | Study DF: **Ophthalmic Solution**

[Study used eye drops \(ophthalmic solution\)](#)

Source: Taylor & Francis

Reason: Dosage form matches product.

Entry #8 SAFETY

Title: The effect of cold tetracaine on the severity of burning sensation upon instillation

Authors: Wiwan Sansanayudh, Thitima Phansucharitthai & Nakarin Sansanayudh

DOI/URL: <https://doi.org/10.2147/OPTH.S179794>

✓ COMPLIANT C2 — Article Title Verification

<https://doi.org/10.2147/OPTH.S179794>

Source: Dove Medical Press

Reason: Title matches publication at DOI.

✓ COMPLIANT C3 — Author Verification

[Authors match publication](#)

Source: Dove Medical Press

Reason: Authors match publication.

✓ COMPLIANT C4 — DOI Resolution

[Resolves to correct publication](#)

Source: Dove Medical Press

Reason: DOI resolves to correct publication.

✓ COMPLIANT C7 — Reference Type

Type: **Clinical Study**

Reason: Reference is a clinical study.

✓ COMPLIANT C8 — Therapeutic Alignment

SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac.

Reason: Reference is relevant to the approved indication.

✓ COMPLIANT C13 — Dosage Form Alignment

Product DF: **Ophthalmic Solution** | Study DF: **Ophthalmic Solution**

[Study used ophthalmic solution](#)

Source: Dove Medical Press

Reason: Dosage form matches product.

Entry #9 EFFICACY

Title: Prospective, randomized, contralateral eye comparison of tetracaine and proparacaine for pain control in laser in situ keratomileusis and photorefractive keratectomy
Authors: Majid Moshirfar, Mark D Mifflin, Michael V McCaughey & Adam J Gess
DOI/URL: <https://doi.org/10.2147/OPTH.S66701>

✔ COMPLIANT C2 — Article Title Verification https://doi.org/10.2147/OPTH.S66701 Source: Dove Medical Press Reason: Title matches publication at DOI.
✔ COMPLIANT C3 — Author Verification Authors match publication Source: Dove Medical Press Reason: Authors match publication.
✔ COMPLIANT C4 — DOI Resolution Resolves to correct publication Source: Dove Medical Press Reason: DOI resolves to correct publication.
✔ COMPLIANT C7 — Reference Type Type: RCT Reason: Reference is a randomized controlled trial.
✔ COMPLIANT C8 — Therapeutic Alignment SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac. Reason: Reference is relevant to the approved indication.
✔ COMPLIANT C13 — Dosage Form Alignment Product DF: Ophthalmic Solution Study DF: Ophthalmic Solution Study used ophthalmic solution Source: Dove Medical Press Reason: Dosage form matches product.

Entry #10 WRONG DF

Title: INTRAVENOUS ADMINISTRATION OF TETRACAINE (PONTOCAINE®) HYDROCHLORIDE Preliminary Report
Authors: J. S. HORAN, M.D.
DOI/URL: [doi:10.1001/archinte.1950.00230120081008](https://doi.org/10.1001/archinte.1950.00230120081008)

✔ COMPLIANT C2 — Article Title Verification https://jamanetwork.com/journals/jamainternalmedicine/article-abstract/554964 Source: JAMA Network Reason: Title matches publication at DOI.
✔ COMPLIANT C3 — Author Verification J. S. HORAN, M.D. Source: JAMA Network Reason: Author matches publication.
✔ COMPLIANT C4 — DOI Resolution Resolves to correct publication Source: JAMA Network Reason: DOI resolves to correct publication.
✔ COMPLIANT C7 — Reference Type Type: Case Report Reason: Reference is a case report.
✗ NON-COMPLIANT C8 — Therapeutic Alignment SmPC Indication: Ocular anaesthetic for topical instillation into the conjunctival sac. Reason: Reference is not relevant to the approved indication (IV use, not ophthalmic).
✗ NON-COMPLIANT C13 — Dosage Form Alignment Product DF: Ophthalmic Solution Study DF: Intravenous Solution Study used intravenous administration Source: JAMA Network Reason: Dosage form does not match product.

Literature Coverage Checks

✖ NON-COMPLIANT

COV1 — PK Coverage (≥1 PK reference)

Count: 0 references | Entries: none

Reason: No PK/PD study identified among the 10 references.

✔ COMPLIANT

COV2 — Efficacy Coverage (≥1 clinical study)

Count: 6 references | Entries: 2, 3, 4, 6, 7, 9

Reason: Multiple clinical efficacy studies present.

✔ COMPLIANT

COV3 — Safety Coverage (≥1 safety reference)

Count: 2 references | Entries: 5, 8

Reason: At least one safety reference present.

⚠ INCONCLUSIVE

COV4 — Pediatric Coverage (conditional)

Count: 0 references | Entries: none

Reason: No explicit pediatric study identified; SMPC allows use in children but no specific pediatric study cited.

❓ SKIPPED

COV5 — Combination Coverage (conditional)

Count: 0 references | Entries: none

Reason: Product is not a combination.

D. Bibliography Section

⚠ INCONCLUSIVE

D1 — Bibliography Section Header

Reason: No explicit bibliography section header in DOC1.

Arithmetic Verification

Check	Expected	Calculated	Status
Entry Serial Number Sequence	1-10 sequential, no gaps/duplicates	1-10 sequential	✔ COMPLIANT

Cross-Document Mismatches

Property	Doc Value	Ref Doc	Ref Value	Web Value	Impact
Reference 10 Dosage Form	Intravenous administration	DOC2(SmPC)/DOC3(Spec-FP)	Ophthalmic Solution	Intravenous Solution	Critical — Reference 10 does not match product dosage form or indication.

Recommended Actions

- CRITICAL

Action 1: Reference 10 is not relevant to the product's approved indication or dosage form.
Error: "INTRAVENOUS ADMINISTRATION OF TETRACAINE (PONTOCAINE®) HYDROCHLORIDE" → Correct: "Replace with a reference using ophthalmic solution for ocular anesthesia."
Supporting: DOC2, DOC3
- MAJOR

Action 2: No PK/PD study among references.
Error: "No PK/PD reference" → Correct: "Add at least one PK/PD study using ophthalmic solution."
Supporting: DOC2, DOC3

Inconclusive Resolutions

- ⚠

Explicit bibliography section in DOC1 — Unable to verify bibliography-table consistency.
Affected (1): D1

Reviewer Remarks

- Reference 10 must be replaced with a study using ophthalmic solution for ocular anesthesia.
- At least one PK/PD study using ophthalmic solution must be added.
- Otherwise, table structure and cross-document identity fields are compliant.
- No explicit bibliography section detected in DOC1.